THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The decimal value of 175_8 is: 1

(a) 125 **(b)** 123 **(c)** 130 **(d)** 127

2. The angles of a quadrilateral are in the ratio 2:3:4:6. The largest angle is: 1

(a) 96° **(b)** 144° **(c)** 132° **(d)** 156°

3. The sum of three odd numbers is: 1

(a) odd **(b)** always prime **(c)** a multiple of 3 **(d)** even

4. $(x+4)^2$ equals: 1

(a) x^2+8x **(b)** $x^2+4x+16$ **(c)** x^2+16 **(d)** $x^2+8x+16$

5. The ratio of 1 m 50 cm to 30 cm is: 1

(a) 5 : 1 **(b)** 1 : 5 **(c)** 6 : 1 **(d)** 5 : 2

6. 35 is what percent of 140? 1

(a) 50% **(b)** 25% **(c)** 22% **(d)** 28%

7. The legs of a right triangle are 2 cm and 6 cm. Its hypotenuse (in cm) is: 1

(a) $\sqrt{41}$ **(b)** 8 **(c)** $2\sqrt{10}$ **(d)** $2\sqrt{2}$

8. If A:B = 3:4 and B:C = 5:6, then A:B:C is: 1

(a) 3:9:6 **(b)** 3:4:6 **(c)** 15:24:10 **(d)** 15:20:24

9. A Koch curve starts with one segment. The number of segments at stage 2 is: 1

(a) 8 **(b)** 16 **(c)** 15 **(d)** 9

10. The mean of 6 numbers is 25. If 5 of them are 30, 22, 28, 20, 24, the remaining number is: 1

(a) 24 **(b)** 31 **(c)** 26 **(d)** 25

11. The sum of 4 consecutive integers starting with n is: 1

(a) $4n$ **(b)** $4n+7$ **(c)** $n+6$ **(d)** $4n+6$

12. The area of a triangle is 130 cm² and its base is 20 cm. Its height (in cm) is: 1

(a) 13 **(b)** 16 **(c)** 15 **(d)** 26

13. The triangular number T_{10} (sum of 1 to 10) is: 1

(a) 100 **(b)** 55 **(c)** 65 **(d)** 45

14. The value of $(-2)^5$ is: 1

(a) 10 **(b)** 32 **(c)** 37 **(d)** -32

15. The number 150 written in base 8 is: 1

(a) 227 **(b)** 225 **(c)** 226 **(d)** 622

16. In a parallelogram ABCD, $\angle A = (3x + 5)^\circ$ and $\angle C = (4x - 10)^\circ$. The value of x is: 1

(a) 20 **(b)** 14 **(c)** 16 **(d)** 15

17. Which of the following numbers is divisible by 11? 1

(a) 8240 **(b)** 8241 **(c)** 8239 **(d)** 8238

18. $(x - 5)^2$ equals: 1

(a) $x^2 + 10x + 25$ **(b)** $x^2 - 10x + 25$ **(c)** $x^2 - 25$ **(d)** $x^2 - 5x + 25$

19. If A : B = 3 : 4 and B : C = 8 : 7, then A : B : C is: 1

(a) 3 : 4 : 7 **(b)** 3 : 8 : 7 **(c)** 6 : 8 : 7 **(d)** 3 : 4 : 56

20. A cycle bought for ₹ 2400 is sold for ₹ 2160. The loss percent is: 1

(a) 9% **(b)** 10% **(c)** 11% **(d)** 20%

21. The legs of a right triangle are 11 cm and 13 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 143 **(b)** 290 **(c)** 24 **(d)** 576

22. 20 men can do a piece of work in 9 days. 15 men will do it in (days): 1

(a) 12 **(b)** 10 **(c)** 180 **(d)** 15

23. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 3 is: 1

(a) $\frac{26}{64}$ **(b)** $\frac{9}{12}$ **(c)** $\frac{1}{64}$ **(d)** $\frac{27}{64}$

24. The mean of 5 observations 11, 27, 22, 19, 31 is 22. If a new observation 70 is added, the new mean is: 1

(a) 22 **(b)** 25 **(c)** 30 **(d)** 92

25. In a number pyramid each block is the sum of the two below it. The bottom row is 2, 5, 1, 3. The top block is: 1

(a) 11 **(b)** 17 **(c)** 22 **(d)** 23

26. In a quadrilateral, a diagonal of 24 cm has perpendiculars of 5 cm and 7 cm drawn to it from the opposite vertices. Its area is: 1

(a) 288 cm² **(b)** 60 cm² **(c)** 144 cm² **(d)** 36 cm²

27. The sum of the consecutive triangular numbers T_9 and T_{10} is: 1

(a) 100 **(b)** 81 **(c)** 110 **(d)** 101

28. The standard form of 384400 is: 1

(a) 38.44×10^5 **(b)** 3.844×10^4 **(c)** 3.844×10^6 **(d)** 3.844×10^5

29. A Babylonian numeral has 3 group(s) of 60 and 40 units. Its value is: 1

(a) 140 **(b)** 58 **(c)** 43 **(d)** 220

30. In rhombus ABCD, $\angle ABC = 100^\circ$. The angle $\angle OBC$ (O is the point where the diagonals meet) is: 1

(a) 80° **(b)** 100° **(c)** 40° **(d)** 50°

31. Which of the following numbers is divisible by 6? 1

(a) 6152 **(b)** 6150 **(c)** 6149 **(d)** 6151

32. $(x + 6)(x - 6)$ equals: 1

(a) $x^2 - 12x + 36$ **(b)** $x^2 + 36$ **(c)** $x^2 + 12x - 36$ **(d)** $x^2 - 36$

33. On a map, 1 cm represents 12 km. A distance of 4 cm on the map is actually: 1

(a) 16 km **(b)** 48 km **(c)** 60 km **(d)** 4.8 km

34. The marked price of a jacket is ₹ 1800 and it is sold for ₹ 1620. The discount percent is: 1

(a) 11% **(b)** 20% **(c)** 10% **(d)** 9%

35. Taking $m = 9$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 18, 81, 82 **(b)** 18, 80, 82 **(c)** 18, 80, 83 **(d)** 9, 80, 82

36. On a map the scale is 1 cm = 25 km. Two places 175 km apart are shown on the map at a distance (in cm) of: 1

(a) 4375 **(b)** 7 **(c)** 8 **(d)** 150

37. A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 24 **(b)** 96 **(c)** 8 **(d)** 25

38. A plant's height (cm) at the end of weeks 1 to 6 is 6, 9, 11, 17, 17, 17. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 3 and week 4 **(b)** Between week 2 and week 3
(c) Between week 4 and week 5 **(d)** Between week 1 and week 2

39. In a calendar the nine dates in a 3×3 block have the middle date 12. Their sum is: 1

(a) 36 **(b)** 108 **(c)** 117 **(d)** 99

40. The cost of flooring a 25 m by 10 m hall at ₹ 60 per square metre is (in ₹): 1

(a) 310 **(b)** 2100 **(c)** 15000 **(d)** 4200

41. For m = 5, the Pythagorean triplet $(2m,\ m^2 - 1,\ m^2 + 1)$ is: 1

(a) 5, 24, 26 **(b)** 10, 24, 26 **(c)** 10, 23, 27 **(d)** 10, 26, 27

42. Which of the following equals 5.62×10^4? 1

(a) 5620 **(b)** 56200 **(c)** 562000 **(d)** 0.000562

43. A Maya numeral has 5 in the 20s place and 7 in the units place. Its value is: 1

(a) 57 **(b)** 107 **(c)** 93 **(d)** 12

44. The diagonals of a rectangle ABCD meet at O and $\angle AOB = 100^\circ$. Then $\angle OAB$ is: 1

(a) 50° **(b)** 40° **(c)** 100° **(d)** 80°

45. For the number 4721, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 6264 **(b)** 6183 **(c)** 6174 **(d)** 6084

46. $(x + 3)(x + 7)$ equals: 1

(a) $x^2 + 10x + 10$ **(b)** $x^2 + 10x + 21$ **(c)** $x^2 + 21x + 10$ **(d)** $x^2 + 21$

47. 12 workers can finish a job in 20 days. 16 workers will finish it in: 1

(a) 26.666666666666666666666667 days **(b)** 16 days
(c) 17 days **(d)** 15 days

48. The price of an article is ₹ 1500 and GST is charged at 12%. The bill amount (in ₹) is: 1

(a) 1512 **(b)** 1680 **(c)** 1848 **(d)** 3360

49. A triangle has sides 11 cm, 25 cm and 30 cm. It is: 1

(a) not right-angled **(b)** right-angled **(c)** equilateral **(d)** cannot be said

50. In a survey of 45 students, 18 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 288 **(b)** 144 **(c)** 54 **(d)** 154

END OF QUESTION PAPER **Total: 50 Marks**

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